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Boston, Massachusetts 02119
P: (617) 445-3500
F: (617) 541-2128

Project: 226052 DFCI-BIDMC New Cancer Hospital - 1 Joslin Place - Construction
1 Joslin Place
Boston, Massachusetts 02215

RFI #0-PB-53: EV Substation Duplex Main Configuration

Revision	0	**Status**	Closed on 11/24/25
To	Rob Morsi (Suffolk Construction) *(Response Required)*	**From**	Nathan Crowley (Suffolk Construction) 65 Allerton St. Boston, Massachusetts 02119
Date Initiated	Nov 14, 2025	**Due Date**	Nov 21, 2025
Location		**Project Stage**	Bidding
Cost Impact		**Schedule Impact**	
Spec Section		**Cost Code**	
Drawing Number	E6.01, E6.02, E7.11, E7.22	**Reference**	Electrical
Linked Drawings			
Received From		**Sub Job**	
Copies To			

ADDITIONAL FIELDS

Required Reviewers	**Urgency**

Activity

Question

Question from Nathan Crowley Suffolk Construction *on Friday, Nov 14, 2025 at 11:48 AM EST*

The EV substation is shown single ended with a duplex main with auto transfer. The second 13.8 kV circuit provides redundancy in the event of a primary cable fault. However, the substation serves non-critical EV charger loads, and the perceived likelihood of a primary cable fault is low.

Please advise if the substation configuration should be changed to a single primary, single ended substation. If yes, please advise if the 15kV cubicle that serves the second primary should be removed from the scope of work. There are currently (10) spare 15kV circuit breakers and there will be (9) spare breakers if one cubicle is eliminated.
Alternatively, please advise if the substation should be priced as shown on the infrastructure drawing set, with a deduct alternate option for consideration by owner.

Attachments
EV Substation Duplex Main.pdf

Official Response

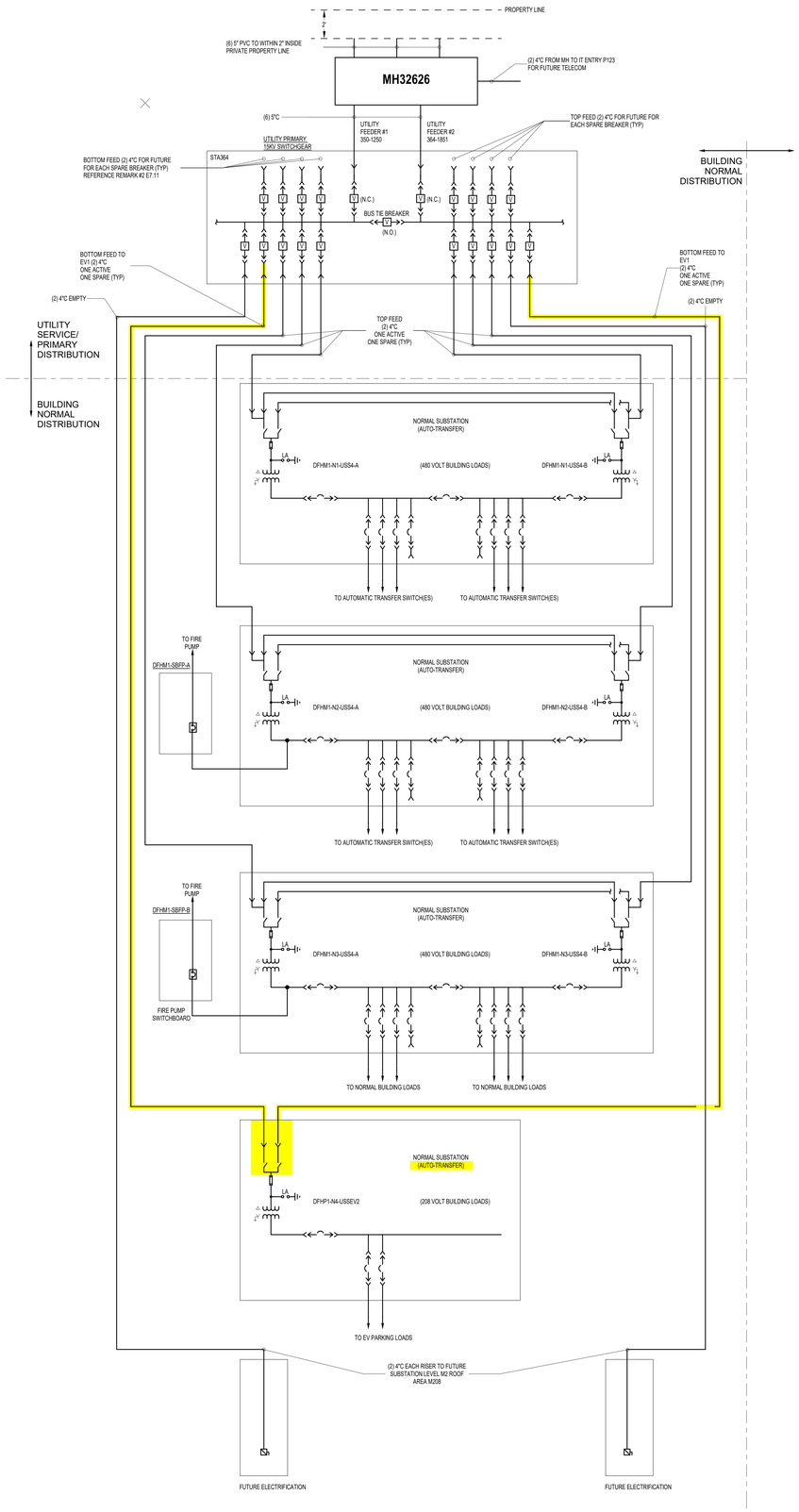
Response from Henry Sniezek Suffolk Construction *on Monday, Nov 24, 2025 at 01:54 PM EST*

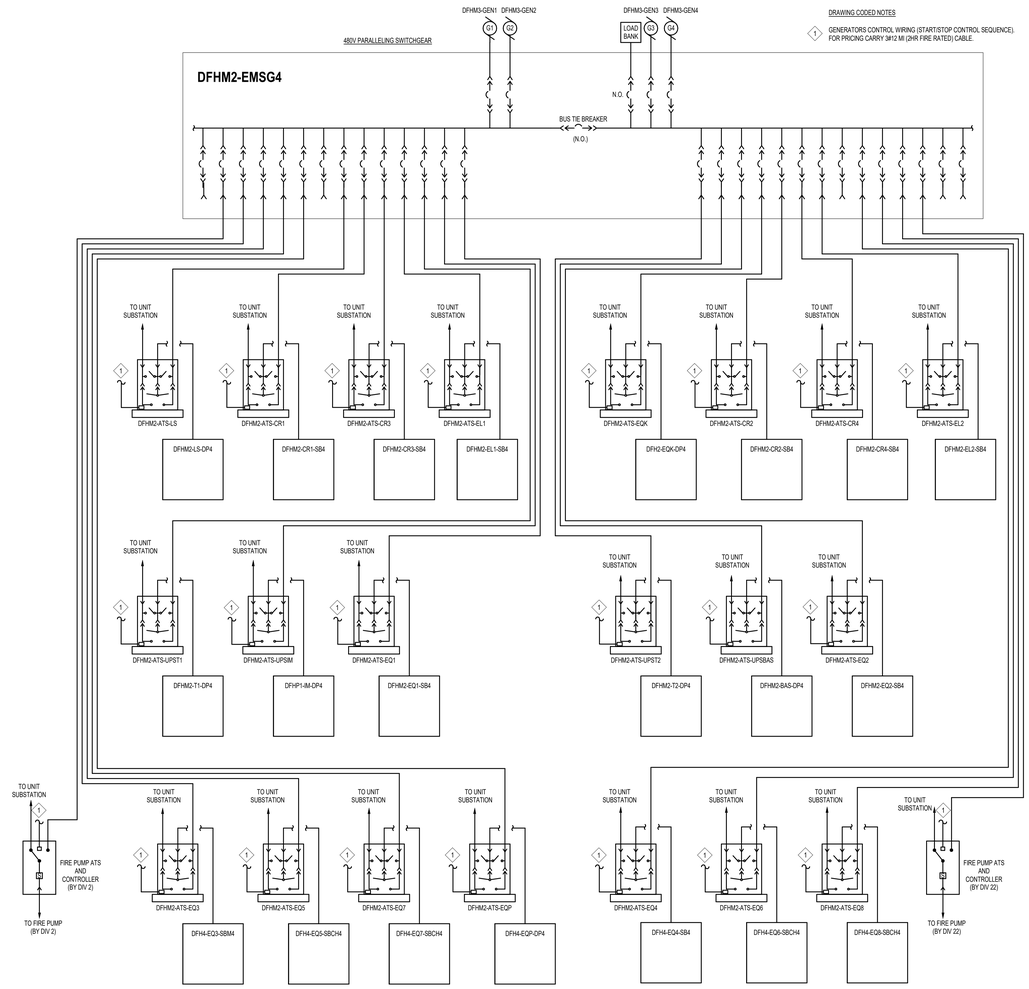
Per Payette 11/21 response: Substation shall be priced as shown on the infrastructure set. A deduct alternate can be provided for consideration. A conversation with ownership is needed to remove the secondary feed from this substation.

All Replies

Response from Henry Sniezek Suffolk Construction *on Monday, Nov 24, 2025 at 01:54 PM EST*

Per Payette 11/21 response: Substation shall be priced as shown on the infrastructure set. A deduct alternate can be provided for consideration. A conversation with ownership is needed to remove the secondary feed from this substation.





DRAWING CODED NOTES

1. GENERATORS CONTROL WIRING (START/STOP CONTROL SEQUENCE). FOR PRICING CARRY 3#12 MI (2HR FIRE RATED) CABLE.

BUILDING INITIAL
FLOOR
SUPPLY SYSTEM
SUPPLY SYSTEM NUMBER
PANEL / EQUIPMENT TYPE
LOAD DESIGNATION
SYSTEM VOLTAGE
EQUIPMENT SERIES IDENTIFIER

DFH 4 - N 1 - SB M 4 - A

BUILDING NAMED.D.:

DFH - DANA FUTURE HOSPITAL

SUPPLY SYSTEM:

"N" - NORMAL POWER
2023 NEC 517.34 "CR" - CRITICAL BRANCH POWER
2023 NEC 517.33 "EL" - ELEVATOR EQUIPMENT BRANCH
2023 NEC 517.35 "EQ" - EQUIPMENT BRANCH
"IM" - IMAGING EQUIPMENT BRANCH
"GEN" - GENERATOR
2023 NEC 517.33 "LS" - LIFE SAFETY POWER
"UPS" - UNINTERRUPTIBLE POWER SYSTEM

PANEL/EQUIPMENT TYPE:

"P" - PANELBOARD
"DP" - DISTRIBUTION PANEL
"SB" - SWITCHBOARD
"SG" - SWITCHGEAR
"USS" - UNIT SUBSTATION
"ATS" - AUTOMATIC TRANSFORM SWITCH

LOAD DESIGNATION:

"BLANK" - MIXED LOADS
"K" - KITCHEN PANEL
"L" - LIGHTING PANEL
"M" - MECHANICAL PANEL
"R" - RECEPTACLE PANEL
"T" - TELECOMMUNICATIONS PANEL
"X" - LAB PANEL
"PH" - PHARMACY
"SL" - SITE LIGHT

SYSTEM VOLTAGE:

"4" - 480/277V
"2" - 208/120V

LEVEL DESIGNATION:

"P1" - LEVEL P1
"1" - LEVEL 1
"2" - LEVEL 2
"M1" - LEVEL M1

ELECTRICAL PANEL NAMING DESIGNATION

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Issues

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Revisions

Number	Date	Description

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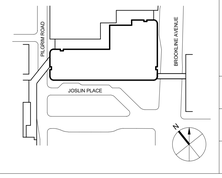

Dana-Farber Cancer Institute
Beth Israel Lahey Health
Beth Israel Deaconess Medical Center

FUTURE CANCER HOSPITAL

1 JOSLIN PLACE, BOSTON MA 02115

Drawing Scale
As indicated

Project Number
17000.00

Date Issued
10/30/2025

ELECTRICAL ONE-LINE RISER DIAGRAM

E6.01

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DRAWING KEY NOTE

1. THE VEHICLE CHARGING POWER PANEL MAIN CIRCUIT BREAKER IS TO BE PROVIDE WITH A SHUNT-TRIP CAPABILITY. PROVIDE A REMOTE "EPO" SWITCH AND CONTROL WIRING TO TURN OFF POWER TO VEHICLE CHARGERS WHEN EPO IS ACTIVATED. COORDINATE WITH THE LOCAL FIRE DEPARTMENT FOR THE "EPO" EXACT LOCATION.
2. GENERATORS CONTROL WIRING (START/STOP CONTROL SEQUENCE). FOR PRICING CARRY 3#12AR (2HR FIRE RATED) CABLE.

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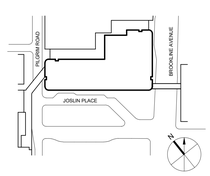

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Beth Israel Deaconess Medical Center

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ELECTRICAL NORMAL RISER DIAGRAM #1

Drawing Scale: 1/8" = 1'-0"
Project Number: 17000.00
Date Issued: 10/30/2025

E6.02

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RELAY SYMBOLS

- ATS — AUTOMATIC TRANSFER SYSTEM
- M — MICROPROCESSOR METERING DEVICE
- 62 — STUCK BREAKER TIMING RELAY
- 50/51 — PHASE OVERCURRENT RELAY
- 50G/51G — GROUND OVERCURRENT RELAY
- 86 — LOCK-OUT RELAY
- 27 — UNDERVOLTAGE RELAY
- 47 — PHASE FAILURE/PHASE REVERSAL RELAY
- FUSE FOR PT PRIMARY AND/OR SECONDARY SIZE PER MANUFACTURER RECOMMENDATIONS
- POTENTIAL TRANSFORMER (P.T.)
- CURRENT TRANSFORMER (C.T.) - SINGLE RATIO *DRCT* DENOTES DUAL RATIO C.T. *MRCT* DENOTES MULTI RATIO C.T.
- (3) — DENOTES QUANTITIES OF DEVICES
- INSTRUMENT TRANSFORMER WIRING
- CONTROL LOGIC
- 87B — DIFFERENTIAL PROTECTIVE RELAY (MAIN BUS)
- 67 / 67N — DIRECTIONAL OVERCURRENT RELAY
- SA — SINGLE ARRESTOR
- V — VACUUM CIRCUIT BREAKER (DRAW-OUT TYPE)
- N.C./N.O. — NORMALLY CLOSED/NORMALLY OPEN
- 5B — DENOTES COMPARTMENT NUMBER IN SWITCHGEAR
- GROUNDED WYE
- OPEN DELTA
- OPEN DELTA WITH SAFETY GROUND

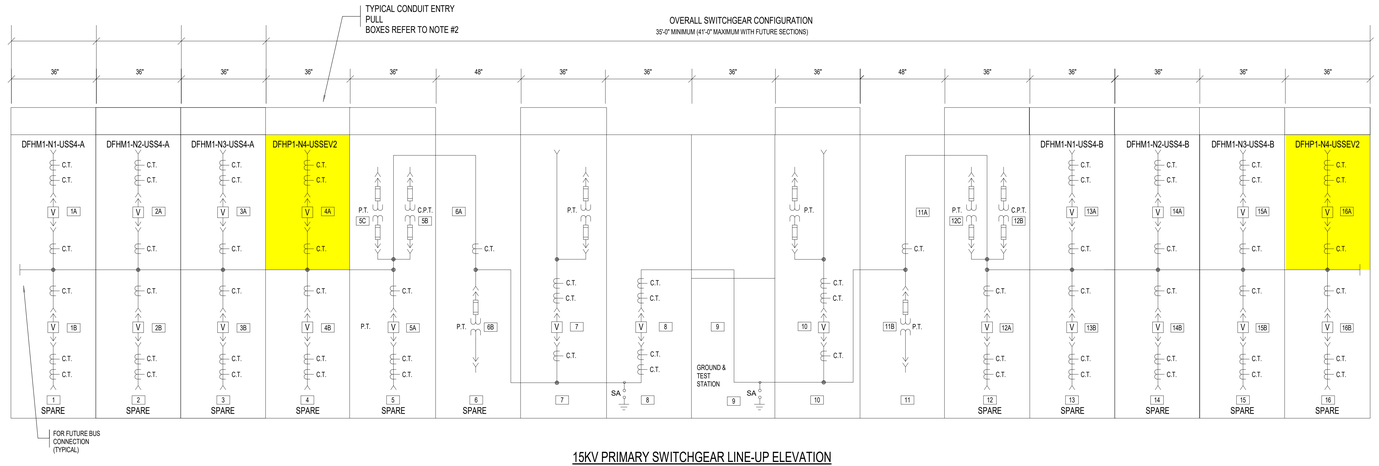
UTILITY 15KV PRIMARY SWITCHGEAR #1 LINE-UP SCHEDULE

HORZ. BUS: 1200A | VERT. BUS: 1200A | S.C.R.: 750MVA | NEUT. BUS: -
GROUND BUS: FULL | NEMA CLASS: * | VOLTAGE: 13.8KV 3PH, 3W | ENCL. NEMA TYPE: *

COMPT. NO.	EQUIPMENT DESIGNATION	DISCONNECT DEVICE				FEEDER						REMARKS
						WIRE		GROUND		CONDUIT		
		POLES	FRAME	TRIP	TYPE	NO.	SIZE	NO.	SIZE	NO.	SIZE	
1A	SUBSTATION 'DFHM1-N1-USS4-A'	3	1200	*	*	3	350 (90°C)	1	40 (600V)	-	-	2
1B	SPARE	3	1200	*	*	-	-	-	-	-	-	3
2A	SUBSTATION 'DFHM1-N2-USS4-A'	3	1200	*	*	3	350 (90°C)	1	40 (600V)	-	-	2
2B	SPARE	3	1200	*	*	-	-	-	-	-	-	3
3A	SUBSTATION 'DFHM1-N3-USS4-A'	3	1200	*	*	3	350 (90°C)	1	40 (600V)	-	-	2
3B	SPARE	3	1200	*	*	-	-	-	-	-	-	3
4A	SUBSTATION 'DFHP1-N4-USSEV2'	3	1200	*	*	3	350 (90°C)	1	40 (600V)	-	-	2
4B	SPARE	3	1200	*	*	-	-	-	-	-	-	3
5A	SPARE	3	1200	*	*	-	-	-	-	-	-	3
5B/5C	BUS/FEEDER P.T.'S/C.P.T.'S	-	-	-	*	-	-	-	-	-	-	
6A	UTILITY METERING C.T.'S	-	-	-	*	-	-	-	-	-	-	1
6B	UTILITY METERING P.T.'S	-	-	-	*	-	-	-	-	-	-	1
7	MAIN-PRIMARY LINE 1	3	1200	*	*	6	500 (90°C)	2	40 (600V)	-	-	1 4
8	TIE	3	1200	*	*	-	-	-	-	-	-	1
9	BUS TRANSITION/GROUND&TEST STATION	-	-	-	*	-	-	-	-	-	-	1
10	MAIN-PRIMARY LINE 2	3	1200	*	*	6	500 (90°C)	2	40 (600V)	-	-	1 4
11A	UTILITY METERING C.T.'S	-	-	-	*	-	-	-	-	-	-	1
11B	UTILITY METERING P.T.'S	-	-	-	-	-	-	-	-	-	-	1
12A	SPARE	3	1200	*	*	3	350 (90°C)	1	40 (600V)	-	-	3
12B/12C	BUS/FEEDER P.T.'S/C.P.T.'S	-	-	-	*	-	-	-	-	-	-	
13A	SUBSTATION 'DFHM1-N1-USS4-B'	3	1200	*	*	3	350 (90°C)	1	40 (600V)	-	-	2
13B	SPARE	3	1200	*	*	3	-	1	-	-	-	3
14A	SUBSTATION 'DFHM1-N2-USS4-B'	3	1200	*	*	3	350 (90°C)	1	40 (600V)	-	-	2
14B	SPARE	3	1200	*	*	3	-	1	-	-	-	3
15A	SUBSTATION 'DFHM1-N3-USS4-B'	3	1200	*	*	3	350 (90°C)	1	40 (600V)	-	-	2
15B	SPARE	3	1200	*	*	3	-	1	-	-	-	3
16A	SUBSTATION 'DFHP1-N4-USSEV2'	3	1200	*	*	3	350 (90°C)	1	40 (600V)	-	-	2
16B	SPARE	3	1200	*	*	3	-	1	-	-	-	3

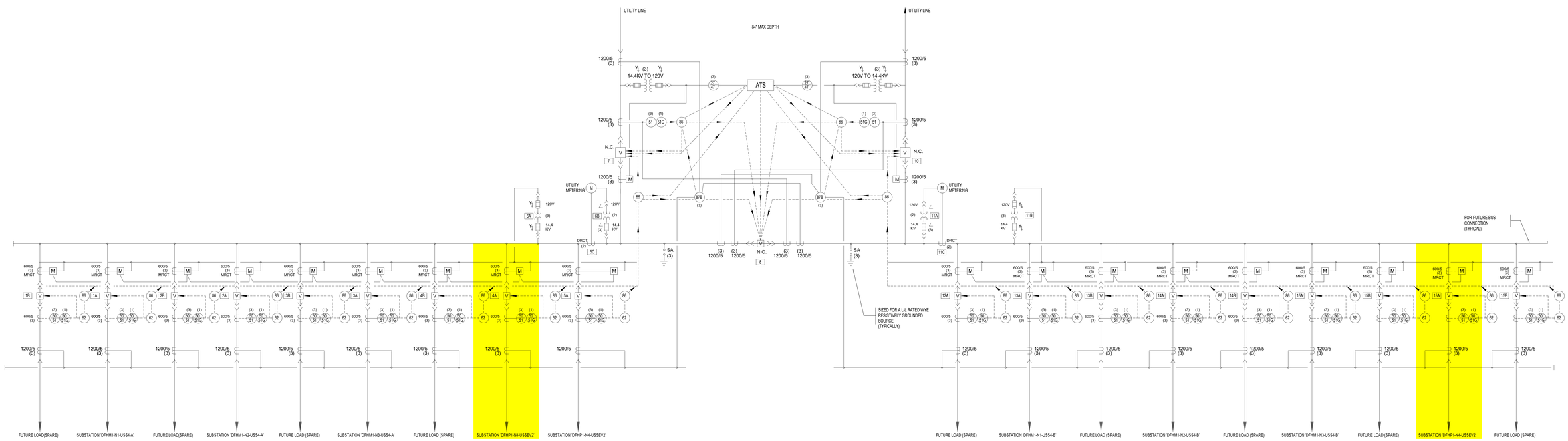
* REFER TO SPECIFICATIONS

1. SWITCHGEAR SHALL CONFORM TO EVERSOURCE REQUIREMENTS FOR METERING CRITERIA AND INCOMING SERVICE.
2. PROVIDE (2)-4" CONDUITS FOR 15KV FEEDERS (ONE ACTIVE, ONE SPARE)
3. PROVIDE (2)-4" CONDUITS FOR SPARE
4. UTILITY "EVERSOURCE" PROVIDING INCOMING PRIMARY FEEDERS FROM MANHOLE IN STREET TO 'STATION 364' PRIMARY SWITCHGEAR. PROVIDE (4)-5" CONDUITS FOR 10KV SERVICE FEEDERS (TWO ACTIVE, TWO SPARE)



15KV PRIMARY SWITCHGEAR LINE-UP ELEVATION

N.T.S.



15KV PRIMARY SWITCHGEAR LINE-UP

ONE LINE DIAGAM

(REFER TO SPECIFICATIONS FOR ADDITIONAL CRITERIA AND REQUIREMENTS)

NOTES:

1. METERING C.T'S & P.T'S, CUBICLES, BUS PHASING AND WIRING SHALL BE IN ACCORDANCE WITH EVERSOURCE CRITERIA-SEE SPECS.

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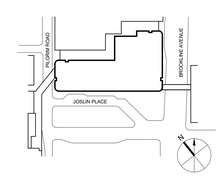

Dana-Farber Cancer Institute | Beth Israel Lahey Health | Beth Israel Deaconess Medical Center

FUTURE CANCER HOSPITAL

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Drawing Scale 1/8" = 1'-0"

Project Number 17000.00

Date Issued 10/30/2025

ELECTRICAL 15KV SWITCHGEAR SCHEDULE#1

E7.11

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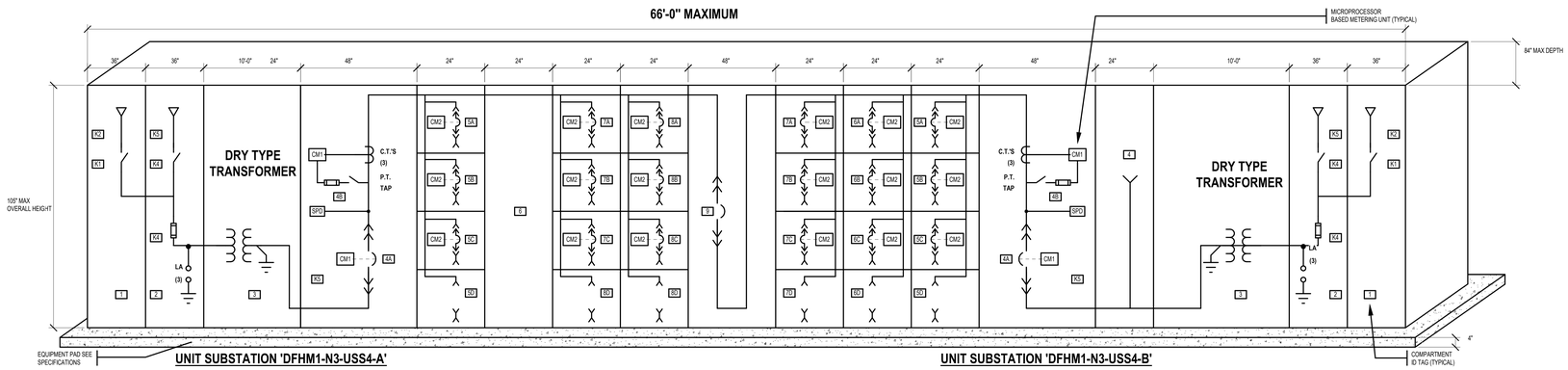


UNIT SUBSTATION 'DFHM1-N3-USS4-A'

UNIT SUBSTATION 'DFHM1-N3-USS4-B'

UNIT SUBSTATION 'DFHM1-N3-USS4-A' SCHEDULE

HORZ. BUS: 4000A	VERT. BUS: 4000A	S.C.R.: 100KAIC	NEUT. BUS: 4000A
GROUND BUS: FULL	NEMA CLASS: SEE SPECS	VOLTAGE: 13.8KV TO 480/277V, 3PH, 4W	ENCL. NEMA TYPE: SEE SPECS

COMPT. NO.	EQUIPMENT DESIGNATION	DISCONNECT DEVICE					FEEDER						REMARKS
							WIRE		GROUND		CONDUIT		
		POLES	FRAME	SENSOR	TRIP	TYPE	NO.	SIZE	NO.	SIZE	NO.	SIZE	
1	PRIMARY DUPLEX SWITCH	3	600	-	-	-	-	-	-	-	-	-	-
2	MAIN FUSIBLE SWITCH	3	600	-	-	200E	SEE SPECIFICATIONS						-
3	2500/3325KVA AA/FA TRANSFORMER	-	-	-	-	-	-	-	-	-	-	-	-
4A	MAIN CIRCUIT BREAKER	3	4,000	4,000	4,000	-	SEE SPECIFICATIONS						(1)
4B	OWNER METERING/TVSS	-	-	-	-	-	-	-	-	-	-	-	-
5A	SWITCHBOARD 'DFHM1-N3-SBL4-A'	3	1,600	1,600	1,600	-	C1600Y	-	-	-	-	-	DFCI LIGHTING (2)
5B	DISTRIBUTION PANEL 'DFHM2-UPST1-DPT4'	3	1,600	1,200	1,200	-	C1200Y	-	-	-	-	-	(VIA 'DFHM2-ATS-UPST1) DFCI (2)
5C	SWITCHBOARD 'DFHI4-N3-SBL4'	3	1,600	1,600	1,600	-	C1600Y	-	-	-	-	-	DFCI LIGHTING (2)
5D	SPACE	3	1,600	-	-	-	-	-	-	-	-	-	BUS & HARDWARE
6	BLANK	3	2,500	-	-	-	-	-	-	-	-	-	-
7A	SWITCHBOARD 'DFHM1-SBR4-A'	3	1,600	1,600	1,600	-	C1600Y	-	-	-	-	-	DFCI RECEPTACLE (2)
7B	SPARE	3	1,600	1,200	1,200	-	-	-	-	-	-	-	(2)
7C	SWITCHBOARD 'DFHM2-CR1-SB4'	3	1,600	1,600	1,600	-	C1600Y	-	-	-	-	-	(VIA 'DFHM2-ATS-CR1) DFCI RECEPTACLE (2)
7D	SPACE	3	1,600	-	-	-	-	-	-	-	-	-	BUS & HARDWARE
8A	SWITCHBOARD 'DFHM1-N3-SBM4-A'	3	1,600	1,600	1,600	-	C1600Y	-	-	-	-	-	DFCI MECHANICAL (2)
8B	SPACE	3	1,600	-	-	-	-	-	-	-	-	-	
8C	SWITCHBOARD 'DFHM2-EQ1-SBM4'	3	1,600	1,600	1,600	-	C1600Y	-	-	-	-	-	(VIA 'DFHM2-ATS-EQ1) DFCI MECHANICAL (2)
8D	SPACE	3	1,600	-	-	-	-	-	-	-	-	-	BUS & HARDWARE
9	TIE - CIRCUIT BREAKER	3	4,000	4,000	4,000	-	-	-	-	-	-	-	(1)

* REFER TO SPECIFICATIONS

(1) MAIN-TIE-MAIN AUTOMATIC TRANSFER CONTROL SYSTEM-PROVIDE REMOTE WALL MOUNTED CONTROL PANEL AND LOCAL CONTROL SWITCHES.

(2) PROVIDE TRIP UNIT METER FOR ASHRAE COMPLIANCE.

UNIT SUBSTATION 'DFHM1-N3-USS4-B' SCHEDULE

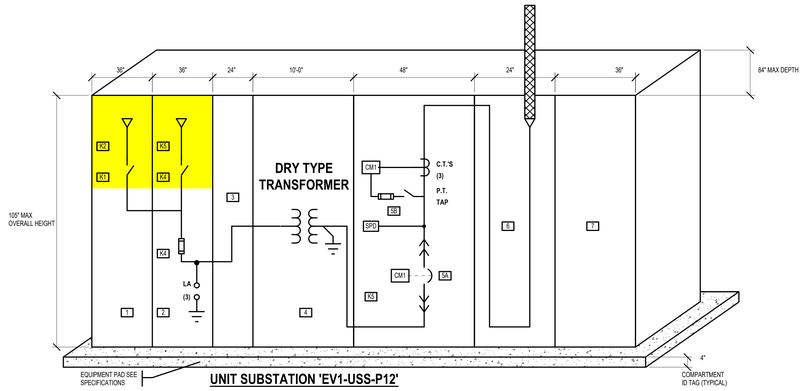
HORZ. BUS: 4000A	VERT. BUS: 4000A	S.C.R.: 100KAIC	NEUT. BUS: 4000A
GROUND BUS: FULL	NEMA CLASS: SEE SPECS	VOLTAGE: 13.8KV TO 480/277V, 3PH, 4W	ENCL. NEMA TYPE: SEE SPECS

COMPT. NO.	EQUIPMENT DESIGNATION	DISCONNECT DEVICE					FEEDER						REMARKS
							WIRE		GROUND		CONDUIT		
		POLES	FRAME	SENSOR	TRIP	TYPE	NO.	SIZE	NO.	SIZE	NO.	SIZE	
1	PRIMARY DUPLEX SWITCH	3	600	-	-	-	-	-	-	-	-	-	-
2	MAIN FUSIBLE SWITCH	3	600	-	200E	-	SEE SPECIFICATIONS						-
3	2500/3325KVA AA/FA TRANSFORMER	-	-	-	-	-	-	-	-	-	-	-	-
4	FIRE PUMP TAP SECTION	3	2,500	-	-	-	(2) SETS OF 4#350KCMIL, 3"C						WIRING SIZE TO FIRE PUMP BREAKER
5A	MAIN CIRCUIT BREAKER	3	4,000	4,000	4,000	-	SEE SPECIFICATIONS						(1)
5B	OWNER METERING/TVSS	-	-	-	-	-	-	-	-	-	-	-	-
6A	SWITCHBOARD 'DFHM1-N3-SBL4-B'	3	1,600	1,600	1,600	-	C1600Y	-	-	-	-	-	DFCI LIGHTING (2)
6B	SPARE	3	1,600	1,200	1,200	-	-	-	-	-	-	-	-
6C	SWITCHBOARD 'DFHI4-N3-SBR4-B'	3	1,600	1,600	1,600	-	C1600Y	-	-	-	-	-	DFCI RECEPTACLE (2)
6D	SPACE	3	1,600	-	-	-	-	-	-	-	-	-	BUS & HARDWARE
7A	SWITCHBOARD 'DFHM1-N3-SBR4-B'	3	1,600	1,600	1,600	-	C1600Y	-	-	-	-	-	DFCI RECEPTACLE (2)
7B	SPARE	3	1,600	1,200	1,200	-	-	-	-	-	-	-	BUS & HARDWARE
7C	SWITCHBOARD 'DFHM2-CR3-SB4'	3	1,600	1,600	1,600	-	C1600Y	-	-	-	-	-	(VIA 'DFHM2-ATS-CR3) DFCI RECEPTACLE (2)
7D	SPACE	3	1,600	-	-	-	-	-	-	-	-	-	BUS & HARDWARE
8A	SWITCHBOARD 'DFHM1-N3-SBM4-B'	3	1,600	1,600	1,600	-	C1600Y	-	-	-	-	-	DFCI MECHANICAL (2)
8B	SPARE	3	1,600	1,600	1,600	-	-	-	-	-	-	-	(2)
8C	SWITCHBOARD 'DFHM2-EQ2-SBM4'	3	1,600	1,600	1,600	-	C1600Y	-	-	-	-	-	(VIA 'DFHM2-ATS-EQ2) DFCI MECHANICAL (2)
8D	PHOTOVOLTAIC CIRCUIT BREAKER	3	250	125	125	-	C125Y	-	-	-	-	-	PV METER (2) (3)

* REFER TO SPECIFICATIONS

(1) MAIN-TIE-MAIN AUTOMATIC TRANSFER CONTROL SYSTEM-PROVIDE REMOTE WALL MOUNTED CONTROL PANEL AND LOCAL CONTROL SWITCHES.

(2) PROVIDE TRIP UNIT METER FOR ASHRAE COMPLIANCE.



UNIT SUBSTATION 'EV1-USS-P12'

UNIT SUBSTATION 'EV1-USS-P12' SCHEDULE

HORZ. BUS: 3000A	VERT. BUS: 3000A	S.C.R.: 100K	NEUT. BUS: 4000A
GROUND BUS: FULL	NEMA CLASS: SEE SPECS	VOLTAGE: 13.8KV TO 208/120V, 3PH, 4W	ENCL. NEMA TYPE: SEE SPECS

COMPT. NO.	EQUIPMENT DESIGNATION	DISCONNECT DEVICE					FEEDER						REMARKS
							WIRE		GROUND		CONDUIT		
		POLES	FRAME	SENSOR	TRIP	TYPE	NO.	SIZE	NO.	SIZE	NO.	SIZE	
1	PRIMARY DUPLEX SWITCH	3	600	-	-	-	-	-	-	-	-	-	-
2	MAIN FUSIBLE SWITCH	3	600	-	100E	-	SEE SPECIFICATIONS						-
3	TRANSITION	3	2,500	-	-	-	-	-	-	-	-	-	-
4	1,000/1,333KVA AA/FA TRANSFORMER	-	-	-	-	-	-	-	-	-	-	-	-
5A	MAIN CIRCUIT BREAKER	3	3,000	3,000	3,000	-	SEE SPECIFICATIONS						(1)
5B	OWNER METERING/TVSS	-	-	-	-	-	-	-	-	-	-	-	-
6	2,500A BUS WAY	-	-	-	-	-	-	-	-	-	-	-	-
7	SPACE	-	-	-	-	-	-	-	-	-	-	-	-

* REFER TO SPECIFICATIONS

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Issues

Number	Date	Description
00	10-30-25	FAÇADE AND INFRASTRUCTURE PACKAGE

Revisions

Number	Date	Description

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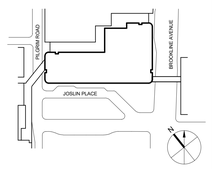

Dana-Farber Cancer Institute
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Beth Israel Deaconess Medical Center

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Drawing Scale
As indicated

Project Number
17000.00

Date Issued
10/30/2025

ELECTRICAL NORMAL SUBSTATION SCHEDULE #2

E7.22

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